at single-cell resolution. In this simulation study, we are interested in testing two aspects of the model.

First, given diverse temporal patterns of gene expression and noisy single-cell measurements, can Sunbear recapitulate single-cell gene expression changes across time? To achieve this goal, we simulated a data set with a relatively large amount of noise (standard deviation of three) and no pseudotime differences among cells (time delay: zero) (Supplemental Methods, "Simulation studies"). We then trained Sunbear on the resulting data set. For the prediction, we started with a single cell's profile from time 0 and varied the time encoding to predict the cell's corresponding profile at continuous time points. We then visualized randomly selected genes' expression changes (Supplemental Fig. S1). Our result suggests that Sunbear is able to capture temporal patterns of gene expression across different cell types and genes.

Next, we are interested in testing whether Sunbear can recapitulate the heterogeneous nature of cell types at single-cell resolution. For each cell type and each time point, we assume that 100 cells have unsynchronized pseudotimes, in which each pair of neighboring cells differs by 0.002 days, and there is a total of 0.2 day differences in pseudotime. We set the standard deviation of noise to one to prevent the noise overriding the signal of cellular differences. In this case, we systematically compared the performance of Sunbear and baseline predictions on each held-out time point, 6.75, 7.25, 7.75, and 8.25, across all genes and cells (Supplemental Methods, "Simulation studies"). Overall, our simulation results suggest that Sunbear can accurately capture individual cell-level differences (Supplemental Fig. S2).

We then set out to test Sunbear's ability to predict a cell's gene expression profile at missing time points. For this analysis, we used a set of 1 million cells randomly sampled from our recently published time-series mouse embryonic development data set, which consists of 12 million cells collected at 67 time points from embryonic day (E) 8 through E18.75 (Fig. 2A; Qiu et al. 2024). To validate Sunbear's performance at cross-time and cross-condition prediction, we held out one time point at a time and trained a model on the remaining time points (see the Methods subsection "scRNA-seq temporal inference evaluation"). In Sunbear, the time point's information is encoded in a latent "time" factor, which is concatenated with the cell identity factor to impute the scRNA-seq profile of each cell. The model's training procedure is designed to encourage the cell identity factors to be invariant of time. Therefore, by fixing the cell factors and varying the time factors, we can predict scRNA-seq profiles in a missing time point.

To quantitatively evaluate Sunbear's prediction performance, we would ideally like to measure Sunbear's ability to predict how a single cell's expression profile changes across time. However, because single cells are destroyed after sequencing, this type of evaluation is impossible. We therefore adopted an alternative approach that relies upon the cell-type annotations created by the authors of the original study. These annotations allow us to ask how well each cell-type-specific pseudobulk profile predicted in the missing time point agrees with the true cell-type-specific pseudobulk profile that is held out from training (see the Methods subsection "scRNA-seq temporal inference evaluation").

Specifically, to validate Sunbear's performance on temporal inference, we asked whether the profile of each cell in the held-out time point can be recapitulated by using cells from the corresponding cell type at a neighboring time point. To do that, we compared Sunbear's prediction (i.e., by varying the time factor of the query cell type at a neighboring time point to the held-out time point) with the baseline prediction, which is the pseudobulk profile of cells in the time point immediately before or after the held-out time point. Even though Sunbear did not see the cell-type information during training, it was able to predict scRNA-seq profiles in an unseen time point better than the baseline

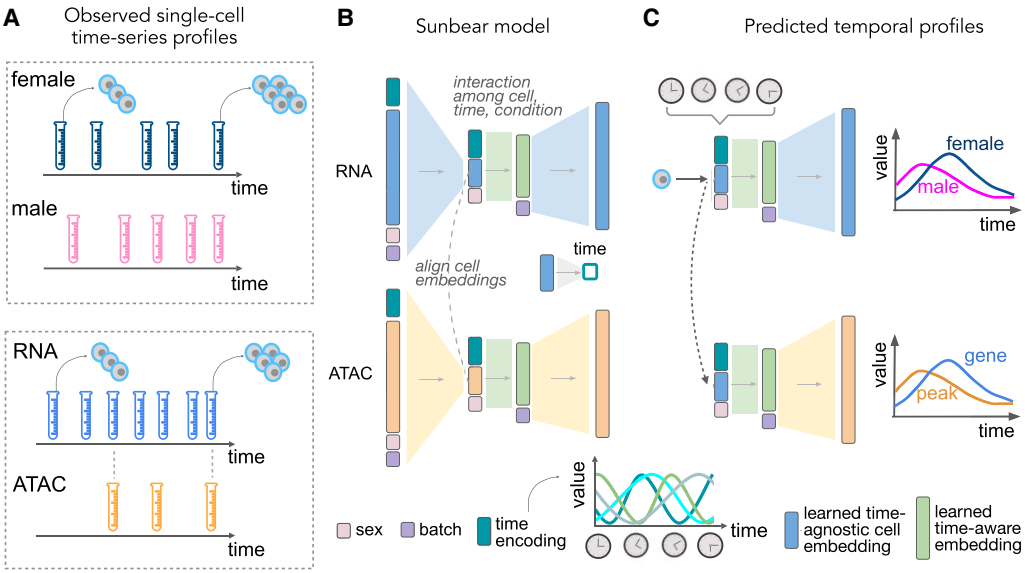


**Figure 1.** Sunbear framework. (*A*) Sunbear takes as input a collection of measurements of single cells at multiple time points in two or more biological conditions (*top*) or data modalities (*bottom*). (*B*) During the training phase, Sunbear learns to decompose the original time-series profiles into four components: cell embedding, time point, batch, and condition. Batch and condition factors are represented by one-hot encodings. The time factor is represented by a sinusoidal encoding. The cell embedding is learned from the original profile and is conditionally independent of the other factors. In the multimodal setting, cell identities are aligned between data modalities. (*C*) In the prediction phase, Sunbear concatenates the query cell's identity factor while varying other factors to impute the cell's profile across time and conditions. By sharing cell embeddings across modalities, Sunbear allows joint temporal modeling of multimodal profiles.